

Supporting Social Movements Through HCI and Design

Adrian Petterson
University of Toronto
Toronto, Canada

Ashique Ali Thuppilikkat
University of Toronto
Toronto, Canada

Paridhi Gupta
University of Zurich
Zurich, Switzerland

Shamika Klassen
University of Colorado Boulder
Boulder, USA
shamika.klassen@colorado.edu

Maggie Jack
Syracuse University / University of
California - Irvine
New York City, USA
mcj46@cornell.edu

Jun Liu
University of Copenhagen
Copenhagen, Denmark
liujun@hum.ku.dk

Priyank Chandra
University of Toronto
Toronto, Canada
priyank.chandra@utoronto.ca

ABSTRACT

The use of digital technologies in grassroots community organizing has multifaceted implications. It has extended the scope of sharing information and experiences, building solidarities and coordination, and fostering common identities to enable participation and amplify the voice of diverse actors in social movements. However, the rise of surveillance technologies, computational propaganda and internet shutdowns are creating novel barriers to democratic action, particularly affecting the participatory parity of marginalized grassroots groups. This one-day hybrid workshop will invite conversations on the complex interrelation between ICTs and social movements and devise ways to support grassroots movements by bringing together HCI researchers, activists and designers. We invite formal position papers to participate in the workshop and encourage participants to ideate and contribute to creating zines that can serve as a helpful resource for supporting grassroots movements.

CCS CONCEPTS

• **Human-centered computing;**

KEYWORDS

Social Movements, Design, Activism

ACM Reference Format:

Adrian Petterson, Ashique Ali Thuppilikkat, Paridhi Gupta, Shamika Klassen, Maggie Jack, Jun Liu, and Priyank Chandra. 2023. Supporting Social Movements Through HCI and Design. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3544549.3573812>

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '23, April 23–28, 2023, Hamburg, Germany
© 2023 Copyright held by the owner/author(s).
ACM ISBN 978-1-4503-9422-2/23/04.
<https://doi.org/10.1145/3544549.3573812>

1 BACKGROUND

Increasingly, digital technologies are serving as the medium of choice for grassroots community organizing around the world [1, 5, 10, 43]. Information and Communication Technologies (ICTs), such as social media and communication applications, have opened up new opportunities for organizing, coordinating, and mobilizing [23, 28]. For marginalized groups in particular, who have not had equitable access to mainstream media platforms, the extensive reach of social media and the rise of user-generated content has allowed a transnational voice [4]. The use of digital technologies for social organizing is contentious, as technology simultaneously empowers grassroots movements as well as institutionalized antagonists such as the state and corporations [33, 34, 41]. Further, some movements which harm marginalized groups can also use the same mechanisms of organizing in the digital space [13, 24]. With the imbalance of power between marginalized community organizers and powerful agents, the design and use of technologies can have a significant impact on whether democratic action succeeds or fails.

Digital technologies already support the needs of community organizing in many ways. Low commitment actions such as signing a digital petition or sharing a social media post lower the barrier for participation in digital organizing [14] and are potentially a gateway to becoming more involved in a movement [18]. Digital spaces also allow activists who face barriers to physical protest to participate, such as disabled activists or diasporic movements [26]. Furthermore, digital tools can be leveraged to facilitate coordination and planning, and as a tool to promote democratic practices and give members a voice [2, 39]. Social media platforms also facilitate emancipatory journalism, which refers to social movements creating their own media coverage to self-represent and better represent their intentions and activities [17, 31], and allow the movement to show their identity as a collective [29]. Additionally, digital spaces can become enclave spaces, where members communicate privately to avoid surveillance or control by antagonists, and maintain an archive of collective memory [7, 38].

However, novel barriers have also emerged, such as increased surveillance, computational propaganda and internet shutdown.

Activists in grassroots social movements are also wary of large digital platforms and potentially harmful digital designs [9, 41]. While online public spaces open up entries for many new members, it also allows antagonistic actors to survey and surveil the actions and intentions of social movements [20, 23]. Sophisticated computational propaganda (e.g. political bots, memes, hashtag spamming and viral digital content) and content moderation and manipulation by powerful antagonists along with internet shutdown have affected participatory parity or a level playing field for democratic deliberations in online spaces, holding profound consequences for social movements, particularly for the communities at the margins. This results in a weakened ability for mobilization and counter-speech [16, 42].

The complex interplay between the affordances and barriers for social organizing in the digital sphere shows the opportunity for supporting democratic and socially progressive action, particularly with marginalized grassroots groups. Our workshop will build on existing HCI and CSCW literature that explores the relationship between ICTs with social movements [22, 23, 32, 40] along with how designers can support social movements [11, 12, 35] and community-led activism [3, 19, 25, 30, 36]. The workshop will also engage with scholarship from related fields such as communication studies and political science that have extensively studied ICT use in social movements and challenges faced by communities [1, 6, 7, 21, 27] and interrogate how it can inform HCI research and design. Furthermore, as scholarship on the use of digital technologies in social movements grows, researchers must look into how to study these movements without creating harm (by exposing social movements, enabling extremist movements, publicizing social movement tactics, etc.) and what ways the researcher or technology designers can help facilitate grassroots organizing in digital spaces. Also, we recognize that terms used in this context can be defined in multiple ways, particularly 'democratic', 'socially progressive', and 'activism'. We keep these definitions open to allow for a diversity of perspectives to be held in conversation with each other. Our workshop aims to explore the following questions:

- (1) What are the challenges faced by social movements and community-based activism (eg. computational propaganda, asymmetric online counter-speech), especially in non-Western settings?
- (2) What role can HCI researchers and designers play in addressing these challenges?
- (3) How can researchers, activists and designers contribute to a repository of resources to support community-based activism?
- (4) How can designers evaluate the risks of researching and enabling community-based activism and social movements, to support socially progressive action and not create harm?

We expect to expand the design implications of ICTs in understanding 21st-century workings of – and challenges to – participatory parity, online counter-speech, mobilizations and democracy.

This workshop will bring together scholars studying social movements, enclave spaces, counterpublics, and resistance in the digital sphere. We will collaborate with workshop participants to discuss support mechanisms for academics, activists, and technology designers to facilitate social movements. This work will build on past

CHI Conference workshops that worked to support democratic actions in digital spaces [44] and workshops that created tools to practice politically progressive HCI research [37]. We will similarly work to provide tools for future scholarly work with grassroots community organizing and for the future design of digital technologies to support grassroots movements.

Acknowledging the political spirit of online organizing and the sociopolitical implications of technology design, the output of this workshop will be a zine detailing tactics of support. Zines have historically been a chosen medium for non-mainstream movements to establish counter memories, self-images that go against the representation of the mainstream media [8]. Similar to emancipatory journalism, counter memories allow grassroots organizers to establish cohesive coalition identities. The crafted nature of zines plays into the embodied and personal nature of activism, and allows creators to use artistic elements from their own locality [15]. Prior workshop outputs have demonstrated the value of zines as counter-media, for ex. anticolonial and antiracist HCI [37]. Our workshop will echo this work, and additionally create a space where not only zines from the workshop, but future contributions can be archived for reference.

2 ORGANIZERS

The organizers for this workshop study social movements and community-based activism from a diverse range of perspectives and from different parts of the world:

- **Adrian Pettersson** is a PhD student at the Faculty of Information, University of Toronto. They study design equity, disability justice in the academic institution, and social movements. They are also the node facilitator for Design Justice Network, Toronto. Adrian works as a grassroots community organizer with and for students with access barriers at the university, and speaks on issues of ethics and inclusion in design. Adrian holds a M.I. in User Experience Design from the Faculty of Information, University of Toronto.
- **Ashique Ali Thuppilikkat** is a PhD student at the Faculty of Information, University of Toronto, Canada. He studies the role of technology in social movements and the resistance practices of communities at the margins. In the past, he pursued M.Phil in Political Science from the Centre for Political Studies, Jawaharlal Nehru University, New Delhi.
- **Paridhi Gupta** is a Postdoctoral Fellow in the Department of Social Anthropology and Cultural Studies at University of Zurich, Switzerland. She has worked extensively on gendered social movements in urban areas while focusing on cultural forms of protests. She is interested in mobilization practices within contemporary social movements, and centering activists' voices. Alongside it, she also works in areas of public feminist art, inclusive pedagogies, digital humanities and human geography. She has a PhD in Gender Studies from Jawaharlal Nehru University, India.
- **Shamika Klassen** is passionate about people and technology. After graduating from Stanford University with a degree in African and African-American studies in 2011, she served a year with AmeriCorps in NYC. She went on to study technology and ethics by developing technowomanism at Union

Theological Seminary in the city of New York. There, she also created and developed the concept of a Tech Chaplain. She received her Master of Divinity from Union in 2017. She currently attends CU Boulder as a doctoral student in their Information Science department and is studying technology, ethics, and social justice issues.

- **Maggie Jack** (she/her) researches the role of media in post-conflict healing, along with other questions of work and technology in global contexts. Her forthcoming book *Media Ruins* describes Cambodian post-genocide media reconstruction as a form of future building and subtle political action. She is a postdoctoral scholar at Syracuse University/UC Irvine and holds a PhD in Information Science (2020) from Cornell University.
- **Jun Liu** is an award-winning author and Associate Professor in the Department of Communication, the University of Copenhagen, Denmark. His research covers political sociology, political communication, and comparative and computational social science, with a specific focus on political activism, digital communication, and democratization. His work has received awards from the International Communication Association and the American Political Science Association.
- **Priyank Chandra** is an Assistant Professor in the Faculty of Information at the University of Toronto. His research studies the sociotechnical practices of communities living at the margins of society, with a focus on informality and resistance. He holds a PhD in Information from University of Michigan, Ann Arbor.

3 PRE-WORKSHOP PLANS

Participant recruitment will occur through social media, community outreach, and a workshop website. The workshop team will create a website detailing the purpose and scope of the workshop, a call to action, and a submission portal. The website will be home to a resource library of pre-existing zines from grassroots organizations. We will distribute the website, along with a formal call for participation, to affiliated institutions and faculties to distribute among their members who may be interested in our workshop. We will encourage them to post the call on their social media sites. We will also personally make posts on social media sites such as Twitter, Facebook, and Instagram.

For participation, we will request formal position papers that are 2–4 pages long and adhere to general ACM formatting and CHI standards. We will prioritize papers that are relevant to the key themes of the workshop: supporting grassroots movements, marginalized groups, sociopolitical activism, resistance, and transnational movements. Workshop organizers will then independently review paper submissions, and determine participants collectively over Zoom. We will aim to recruit 10–15 participants.

To provide supporting information prior to the workshop, materials will be posted to the workshop website in advance. The workshop website can be found at <https://sites.google.com/view/hcisocialmovements/>

These materials will include information about zines and how to make them along with copies of zines from other social organizing

efforts. Reference lists for information about social movements and digital spaces will also be posted. Moreover, we will post a list of community discussion guidelines open for comment by participants or other community members. Prior to the conference, we will adjust the community guidelines in response to input. In addition to our website, we will ensure the workshop is a welcoming and productive place through several efforts. We will distribute a survey to all participants clarifying how they plan on attending (online or in person), any accessibility or safety needs, as well as names and pronouns. Along with the survey, we will distribute all necessary guiding materials for the workshop and example materials the participants may bring with them to design and create zines. Prior to the workshop, we will make available to participants the position papers, participant bios, the agenda, and post-workshop plans. We will additionally host introductory Zoom conversations between participants and organizers to clarify questions.

4 WORKSHOP STRUCTURE

The one-day workshop will be delivered in a hybrid online/offline format. We will have a team of organizers operating on-site at the CHI 2023 conference, as well as a team maintaining an online space, which will receive a live stream of the in-person workshop. The reason for this format choice is to increase the accessibility of the workshop. A hybrid format will also allow participants from around the world to participate regardless of the availability of travel funds. This is particularly important due to the inclusion of those unable to travel for the conference due to geographical distance, resource limitations, or other barriers.

We will have 10–15 participants. This smaller number will foster the feeling of enclave space we seek to create in our workshop. Herein, the group negotiates in-group dynamics with insider members, in this case, scholars working to support grassroots movements. During the workshop housekeeping, we will present a list of community guidelines to create a safer space and foster a sense of cohesion among the group. The group will then offer adjustments or novel guideline suggestions.

The workshop will consist of a morning section in which we will present debates and progress around social movements and digital technologies, and then participants will present position papers and receive peer feedback. The presentations will last 2 minutes, and the participants will then have 3 minutes of question and feedback time. Presenters will use one slide sent to the organizers in advance, to reduce transition time during presentations. During discussion and feedback, participants will be encouraged to seek overlaps between their papers and areas of potential collaboration. After the presentations, there will be a debate on the presentations and background materials between the participants. Following the short debate, there will be a 45-minute lunch break.

For the afternoon portion of the workshop, participants will work together to create zines that serve as guidance for future research and design to support grassroots organizing in digital spaces. Participants may use the materials provided but will also be invited to bring crafting materials to the workshop from home. By using this method, we give a nod to the distributed yet local nature of online grassroots organizing and bring in elements from individual contributors while also creating a single, cohesive stance.

Participants will work in groups of 4-5, partnering with participants noted to have complementary work. The zine creation will occur over the course of one hour, after which the groups will present their zines, and receive feedback from other groups on the limits and potentials of their zine.

After the conclusion of the workshop, participants will be asked to give their zines to the organizers, or may keep their zine and agree to scan and send the digital file of their zine to organizers post-workshop if they would like to continue working on it. The zines will then be added to a new living repository of resources for supporting grassroots movements online.

4.1 Agenda

- **9:00** Introductions and housekeeping
- **9:30** Presentation of debates and background materials
- **10:45** Coffee break
- **11:00** Participant presentations
- **12:15** Coffee break
- **12:30** Group debate
- **1:45** Lunch break
- **2:30** Zine design and construction
- **3:30** Group Feedback
- **4:00** Close, collect zines

5 POST-WORKSHOP PLANS

After the conclusion of the workshop, organizers will digitize the zines produced as a result of the workshop, and add them to the workshop website. The website will then serve as a living repository of zines and other resources for researchers, activists and designers to use to support grassroots movements online. A portal will be available for future contributions to the repository to be submitted on the website, and organizers will add resources submitted that fit with the nature of the repository. To support the growth and use of this repository we will maintain a mailing list for supporting digital grassroots action to update on the status of the repository, calls for resource submissions as well as any novel resources contributed. The mailing list will also be used to encourage future research in this area, as a network of scholarly contacts. Participants will also be encouraged to take the position papers submitted to the workshop and develop them into full papers for future conference submissions. Further, with the permission of participants, organizers plan to craft a paper on the workshop and the effect of the repository on scholarship.

6 CALL FOR PAPERS

This workshop invites papers to delve further into the interrelation between ICTs and social movements in different parts of the world. We focus on the complex underpinnings of this interrelation, such as identity formation, enclave spaces and new forms of on-ground mobilizations to reveal ways that scholars, activists and designers can enable grassroots organizing.

We are accepting 2-4 pages long position papers which expand and iterate on the theme. Submissions should focus on forms of digital activism or the intersection of digitality and social movements, speaking particularly to the innovations and challenges to such forms of organization. We would especially appreciate papers

from diverse geographical and identity locations which explore this terrain from a bottom-up approach, as well as auto-ethnographic pieces which comment on the questions. The papers are expected to adhere to the ACM format.

This is a two-part hybrid workshop spread over six hours. The morning session will be spread over two-time slots where participants will present their work followed by a general discussion on the same. In the afternoon session, we will engage in a participatory hands-on approach to designing materials for social activism. The participants will be divided into breakout groups to discuss and create their physical zines employing the idea of digital activism, its modes, potential and limitations. All participants must register for the workshop and at least one day of the CHI 2023 conference.

Information regarding submission and pre- and post-workshop activities will be made available through our [website](#).

REFERENCES

- [1] Sheetal D Agarwal, Michael L Barthel, Caterina Rost, Alan Borning, W Lance Bennett, and Courtney N Johnson. 2014. Grassroots organizing in the digital age: Considering values and technology in Tea Party and Occupy Wall Street. *Information, Communication & Society* 17, 3 (2014), 326–341.
- [2] Eva Anduiza, Camilo Cristancho, and José M Sabucedo. 2014. Mobilization through online social networks: the political protest of the indignados in Spain. *Information, Communication & Society* 17, 6 (2014), 750–764.
- [3] Mariam Asad and Christopher A Le Dantec. 2015. Illegitimate civic participation: supporting community activists on the ground. In *Proceedings of the 18th ACM Conference on Computer Supported Cooperative Work & Social Computing*. 1694–1703.
- [4] William K Carroll and Robert A Hackett. 2006. Democratic media activism through the lens of social movement theory. *Media, culture & society* 28, 1 (2006), 83–104.
- [5] Victoria Carty and Francisco G Reynoso Barron. 2019. Social movements and new technology: The dynamics of cyber activism in the digital age. In *The Palgrave handbook of social movements, revolution, and social transformation*. Springer, 373–397.
- [6] Manuel Castells, Mireia Fernández-Ardévol, Jack Linchuan Qiu, and Araba Sey. 2007. *Mobile Communication and Society: A Global Perspective Manual*. Number July. Massachusetts Institute of Technology, Cambridge. <https://doi.org/0262033550>
- [7] Karma R Chávez. 2011. Counter-public enclaves and understanding the function of rhetoric in social movement coalition-building. *Communication Quarterly* 59, 1 (2011), 1–18.
- [8] Red Chidgey. 2018. *Feminist afterlives: assemblage memory in activist times*. Springer.
- [9] Sasha Costanza-Chock. 2020. *Design justice: Community-led practices to build the worlds we need*. The MIT Press.
- [10] Débora de Castro Leal, Max Krüger, Kaoru Misaki, David Randall, and Volker Wulf. 2019. Guerilla Warfare and the Use of New (and Some Old) Technology: Lessons from FARC's Armed Struggle in Colombia. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. 1–12.
- [11] Jill P Dimond. 2012. *Feminist HCI for real: Designing technology in support of a social movement*. Georgia Institute of Technology.
- [12] Jill P Dimond, Michaelanne Dye, Daphne LaRose, and Amy S Bruckman. 2013. Hollaback! The role of storytelling online in a social movement organization. In *Proceedings of the 2013 conference on Computer supported cooperative work*. 477–490.
- [13] Sean M Eddington. 2018. The communicative constitution of hate organizations online: A semantic network analysis of "Make America Great Again". *Social Media+ Society* 4, 3 (2018), 2056305118790763.
- [14] Jordana J George and Dorothy E Leidner. 2019. From clicktivism to hacktivism: Understanding digital activism. *Information and Organization* 29, 3 (2019), 100249.
- [15] Betsy Greer. 2014. *Craftivism: The art of craft and activism*. Arsenal Pulp Press.
- [16] Paridhi Gupta, Adrian Petterson, Divyani Motla, and Priyank Chandra. 2022. Ladange, Adange, Jeetange: The Farmers' Movement and Its Virtual Spaces. *Proc. ACM Hum.-Comput. Interact.* 6, CSCW2, Article 446 (nov 2022), 34 pages. <https://doi.org/10.1145/3555547>
- [17] Summer Harlow and Thomas J Johnson. 2011. Overthrowing the protest paradigm? How the New York Times, global voices and Twitter covered the Egyptian revolution. *International journal of Communication* 5, 16 (2011), 1359–1374.
- [18] Hyesun Hwang and Kee-Ok Kim. 2015. Social media as a tool for social movements: The effect of social media use and social capital on intention to participate

- in social movements. *International Journal of Consumer Studies* 39, 5 (2015), 478–488.
- [19] Farnaz Irannejad Bisafar, Lina Itzel Martinez, and Andrea G. Parker. 2018. Social Computing-Driven Activism in Youth Empowerment Organizations: Challenges and Opportunities. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) (CHI '18). Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3173574.3173757>
 - [20] Anastasia Kavada. 2015. Communicating protest movements: The case of occupy. *tripleC: Communication, Capitalism & Critique. Open Access Journal for a Global Sustainable Information Society* 13, 1 (2015).
 - [21] Anastasia Kavada. 2020. Creating the collective: social media, the Occupy Movement and its constitution as a collective actor. In *Protest Technologies and Media Revolutions*. Emerald Publishing Limited.
 - [22] Yong Ming Kow, Yubo Kou, Bryan Semaan, and Waikuen Cheng. 2016. Mediating the undercurrents: Using social media to sustain a social movement. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*. 3883–3894.
 - [23] Yong Ming Kow, Bonnie Nardi, and Wai Kuen Cheng. 2020. Be water: Technologies in the leaderless Anti-ELAB movement in Hong Kong. In *Proceedings of the 2020 CHI Conference on human factors in computing systems*. 1–12.
 - [24] Jack LaViolette and Bernie Hogan. 2019. Using platform signals for distinguishing discourses: The case of men's rights and men's liberation on Reddit. In *Proceedings of the International AAAI Conference on Web and Social Media*, Vol. 13. 323–334.
 - [25] Debora de Castro Leal, Angelika Strohmayer, and Max Krüger. 2021. On activism and academia: Reflecting together and sharing experiences among critical friends. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–18.
 - [26] Hojeong Lee. 2020. *Korean Digital Diaspora Transnational Social Movements and diaspora identity*. Rowman & Littlefield.
 - [27] Nathalie Maréchal. 2016. Automation, algorithms, and politics| when bots tweet: Toward a normative framework for bots on social networking sites (feature). *International Journal of Communication* 10 (2016), 10.
 - [28] Jessica McLean. 2020. 'It's Just Coding': Disability Activism In, and About, Digital Spaces. In *Changing Digital Geographies*. Springer, 229–246.
 - [29] Douglas M McLeod. 2007. News coverage and social protest: How the media's protect paradigm exacerbates social conflict. *J. Disp. RESOL.* (2007), 185.
 - [30] Maria Menendez-Blanco, Pernille Bjorn, and Antonella De Angeli. 2017. Fostering cooperative activism through critical design. In *Proceedings of the 2017 ACM Conference on Computer Supported Cooperative Work and Social Computing*. 618–629.
 - [31] Stefania Milan. 2013. *Social movements and their technologies: Wiring social change*. Springer.
 - [32] Aparna Moitra, Syed Ishtiaque Ahmed, and Priyank Chandra. 2021. Parsing the 'Me' in #MeToo: Sexual Harassment, Social Media, and Justice Infrastructures. *Proceedings of the ACM on Human-Computer Interaction* 5, CSCW1 (2021), 1–34.
 - [33] Adam D Reich. 2017. The organizational trace of an insurgent moment: Occupy Wall Street and New York City's social movement field, 2004 to 2015. *Socius* 3 (2017), 2378023117700651.
 - [34] Pedro Sanches, Vasiliki Tsaknaki, Asreen Rostami, and Barry Brown. 2020. Under Surveillance: Technology Practices of those Monitored by the State. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–13.
 - [35] Saiph Savage, Andres Monroy-Hernandez, and Tobias Höllerer. 2016. Botivist: Calling volunteers to action using online bots. In *Proceedings of the 19th ACM Conference on Computer-Supported Cooperative Work & Social Computing*. 813–822.
 - [36] Rafael ML Silva, Erica Principe Cruz, Daniela K Rosner, Dayton Kelly, Andrés Monroy-Hernández, and Fannie Liu. 2022. Understanding AR Activism: An Interview Study with Creators of Augmented Reality Experiences for Social Change. In *CHI Conference on Human Factors in Computing Systems*. 1–15.
 - [37] Rachel Charlotte Smith, Heike Winschiers-Theophilus, Daria Loi, Rogério Abreu de Paula, Asnath Paula Kambunga, Marly Muudeni Samuel, and Tariq Zaman. 2021. Decolonizing design practices: towards pluriversality. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–5.
 - [38] Catherine R Squires. 2002. Rethinking the black public sphere: An alternative vocabulary for multiple public spheres. *Communication theory* 12, 4 (2002), 446–468.
 - [39] Kate Starbird and Leysia Palen. 2011. "Voluntweeters" self-organizing by digital volunteers in times of crisis. In *Proceedings of the SIGCHI conference on human factors in computing systems*. 1071–1080.
 - [40] Marlon Twyman, Brian C Keegan, and Aaron Shaw. 2017. Black Lives Matter in Wikipedia: Collective memory and collaboration around online social movements. In *Proceedings of the 2017 acm conference on computer supported cooperative work and social computing*. 1400–1412.
 - [41] Julie Uldam. 2018. Social media visibility: challenges to activism. *Media, Culture & Society* 40, 1 (2018), 41–58.
 - [42] Samuel C Woolley and Philip N Howard. 2018. *Computational propaganda: Political parties, politicians, and political manipulation on social media*. Oxford University Press.
 - [43] Volker Wulf, Konstantin Aal, Ibrahim Abu Kteish, Meryem Atam, Kai Schubert, Markus Rohde, George P Yerosus, and David Randall. 2013. Fighting against the wall: Social media use by political activists in a Palestinian village. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. 1979–1988.
 - [44] Johanna Ylipulli and Aale Luusua. 2021. In Search of the Alternative Future: Developing Participatory Digital Citizenship to Address the Crisis of Democracy. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–4.